# SIMATS ENGINEERING

## ASSESSMENT TOOL 1: Short Answer Test

**COURSE NAME:** Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

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**COURSE OUTCOME COVERED**

**CO1:** *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 25**
Question Count: 5

R. Bhargav

**Code of Conduct**

I R. Bhargav (192126022) (Name / Reg. No.)
certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: R. Bhargav

**Assessment Tasks**
**Short Answer Test**

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

**Short Answer Test Rubric**

| Criteria | Weightage | Level 4 - Exemplary | Level 3 - Proficient | Level 2 - Developing | Level 1 - Beginning |
|---|---|---|---|---|---|
| Conceptual Accuracy | 30% | Accurate definitions and precise explanation of all key terms | Mostly accurate with minor gaps | Partial understanding with noticeable errors | Incorrect or irrelevant explanation |
| Coverage of Concepts | 20% | Covers all expected sub-points completely | Covers most expected points | Covers only basic points | Very limited coverage |
| Use of Examples | 15% | Uses highly relevant cloud examples to strengthen the answer | Uses relevant examples with minor mismatch | Uses vague or partially relevant examples | No useful examples |
| Comparison / Differentiation | 20% | Clearly distinguishes concepts with strong logic | Reasonable differentiation with minor overlap | Comparison is weak or incomplete | Fails to differentiate concepts |
| Clarity of Expression | 15% | Clear, organized, and concise answer writing | Generally clear with few issues | Understandable but poorly organized | Difficult to follow |

**Total Marks:**

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# CO1 - AT1 (Short answer)

Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.

Defintion: cloud computing is the delivery of computing services such as servers, storage, databases, networking, and software over the internet.

Four characteristics:

1) on demand self-services: users can access computing resources whenever they need computing resources whenever they need them without contacting the service provider.

2) Broad Network access: cloud services can be accessed through the internet using computers, laptops, tablets and mobile phones.

3) Resource pooling: cloud providers share computing resources among multiple users.

4) Rapid elasticity: Resources can be increased or decreased quickly based on user requirements.

2) Explain the evolution of cloud computing

* Main-frame computers provided centralized computing
* Personal computers made computing widely available
* client-server computing enabled resource sharing
* virtualization allowed multiple virtual machines on one server
* Internet and web services enabled Modern cloud platforms like AWS And Azure.

* Web services: Allowed applications to communicate over the internet.

* Modern cloud platform: AWS and Azure provide computing and storage services.

3) Explain server virtualization and differentiate it from cloud computing.

Server virtualization creates multiple virtual machines on a single physical server.

* Improves hardware resource utilization.
* reduces hardware and maintenance costs
* Supports multiple operating systems
* Makes server management easier.

* virtualization is an enabling technology, while cloud computing delivers resources as services.

example: VMware for virtualization and AWS — for cloud computing]

4) Discuss the role of data centers in delivering cloud services.

A data center contains servers, storage systems, and networking equipment.

* Data storage: stores large amounts of user data
* processing: processes applications and user request
* Networking: connects cloud services to the internet.
* Security: protects data from unauthorized access.
* Reliability: Back up power and cooling help keep services running
* Scalability: Allows resources to increase when demand grows

ex: AWS data centers provides computing and storage services.

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5) IaaS: provides virtual machines, storage, and networking.

PaaS: provides a platform to develop and deploy applications

SaaS: provides ready to use software over the internet.

*aas: provides different IT resources as services.

Advantages: These models reduce the need to purchase and maintain physical hardware.

